Energy Modelling

Renewable Energy 80% by 2030

Future Supply

Contents

Brief

Aim

Refine the assumptions and modelling of a key piece of intelligence provided by external consultants *Aurora* in their report *Accelerating Renewables in Northern Ireland, A high level design of a* . Their report details projections of renewable electricity supply going forward. There is criticism of the assumptions used around pipeline progression and risk weighting the viability of projects currently in the pipeline. Also subtle nuance between observed actual projections to 2030, and variables fixed to set the projection to ensure 80% by 2030 are not well understood.

To add value our work should borrow the bulk of correct work in the report, verify with experts and more clearly present figures.

Background

The climate change act contains the stipulation that Northern Ireland reach 80% of total energy supply being renewable by 2030.

1) This requires determination of **current renewable energy generation.** This is the current, existing, renewable generation which we assume to be constant to 2030.

2) The **pipeline of renewable energy projects forthcoming**, and their generation capacity coming available over the next 5 years from 2025 - 2030.

3) Estimation or determination of the **Total Energy Requirement,** from which we can take 80% as the absolute target figure.

A corollary of this work is that fossil fuel and other non - renewable supply is elastic enough to scale down on the basis of increased renewable supply, thus the pipeline of currently developing non-renewable sources are beyond our scope and not of interest.

5 years is noted to be a short time frame on the scale of energy and specifically renewable project development. A more complete consideration arising from this modelling work would involve future energy capacity estimation from projects that may be anticipated to start planning soon, based on current trends of project application and development.
We assume that all projects pertinent to new 2030 supply are already in the planning or development or commissioning process.

It is noted that the current trajectory is way off the expected trajectory and that the target is unrealistic both with the current pipeline of project capacity at the planning stage or later, and absent any real advance in offshore wind capacity.

Demand and Supply

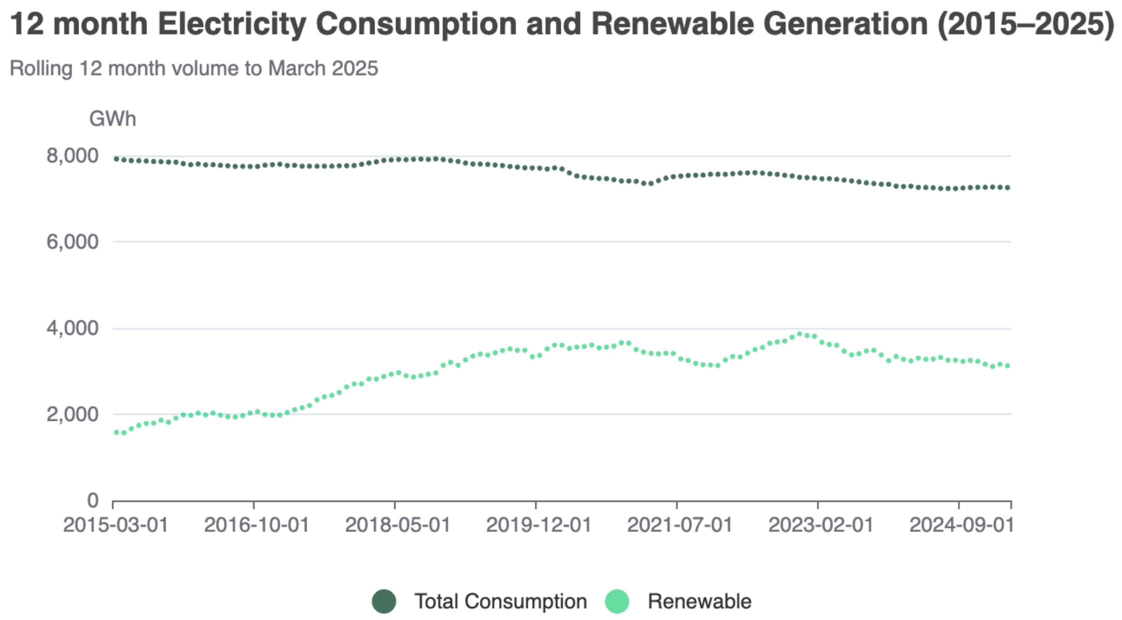
Supply is obviously designed to meet demand. The difference between generation and consumption is more important

Generation is total generated power in the Country.

Consumption is net of transmission and distribution losses, own use, energy exports and other power not consumed in NI, but is inclusive of energy imports.

Non-Renewables
Assumed to be elastic and required only for demand renewables cannot meet. The principle of *priority dispatch* ensures that when renewable sources are available, they are used over non-renewable source.

Renewables
Determining this is the thrust of our modelling work. Future estimates are not well known. SONI and EirGrids All Island Resource Adequacy Assessment gives a project of renewables broken down by type that goes out to 2034, and includes figures that *miss* the 80% by 2030 target.
Past trends of supply are presented below.



The final reported 12 month period is further broken down by renewable type.

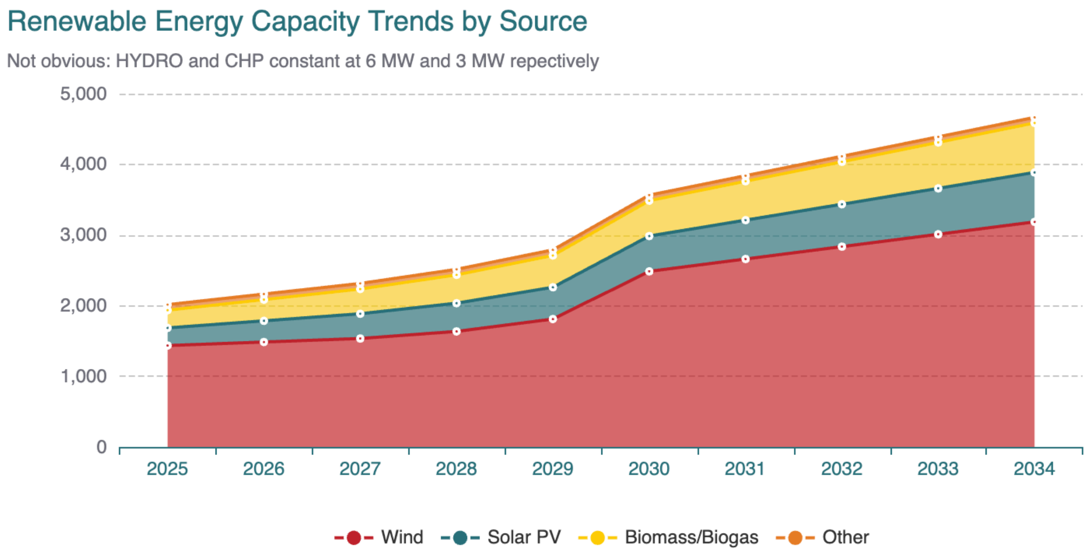
Renewable Energy breakdown

Generation Type	Percentage Contribution	Volume Generated (GWhr)
Biogas	6.8	214
Biomass	5.2	161
Landfill gas	1.6	49
Other	0.9	29
Solar PV	3.6	112
Wind	81.9	2,561
Total	**100**	**3,126**

April 2024 to March 2025

Future Generation and Consumption

SONI also release figures of Renewable energy capacity by source. The plot below is of projections from the All Ireland Adequacy Assessment. It gives forward estimates of renewable supply from 2025 into the future.
The methodology for projecting this is not released.



Key dates are below. There are notable differences that we try and reconcile.

AIEAA projected All - Renewable sources

Year	Renewable capacity (MW)	X * 8760 * CF = 22%) / 1e6 (TWhr)(2)	Adjusted TWhr (see discussion) (2)
2023			
2024			
2025	2,023	3.9	3.1
2030	3,573	6.9	5.5
2034	4,673	9.0	7.2

EirGrid and SONI I All-Island Resource Adequacy Assessment 2025-2034
1) CF = 22%
2) CF = 17.6%

Discrepancy

Notes have been made of possible reasons for the discrepancy.

22% Capacity Factor is a large estimate, 18% worked better

Mismatch between counted sources and the timings 2024-25 financial year vs. 2025 calendar year.

Measured vs. Estimated

The GWh figure may be net of outages, curtailment, own-use or exports, and not gross energy

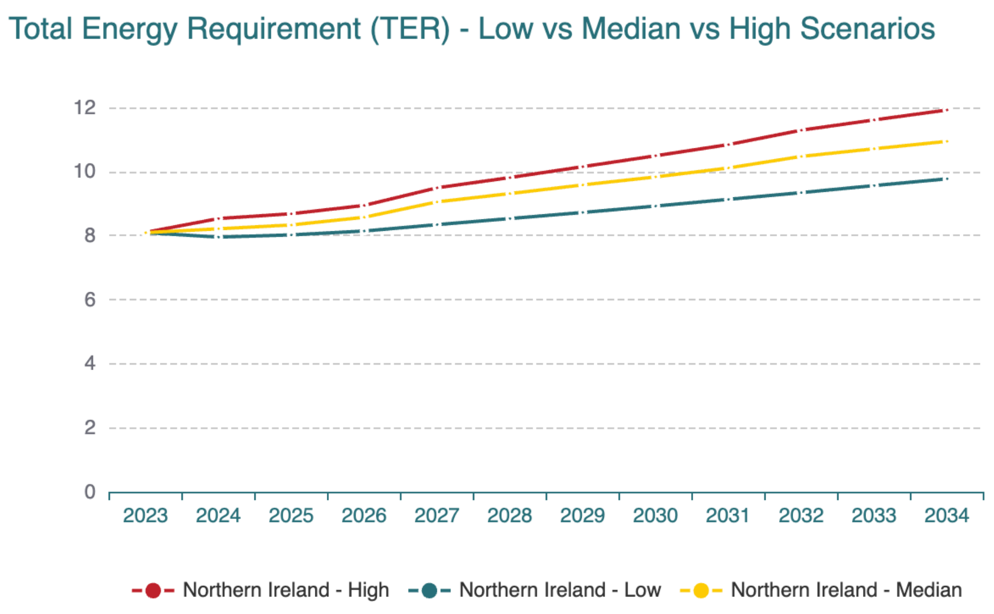
They may be using average capacity over the year rather than nameplate.

Nameplate capacity means the maximum rated output under optimal conditions. Is measured in Watts (GW/MW/TW). I's an instantaneous power measure, not energy over time.

Timing / commissioning profile may not be treated in same across sources, if applied.

Future Demand

Total Energy Requirement is lifted from SONI working paper. It details demand for Northern Ireland out to 2034.



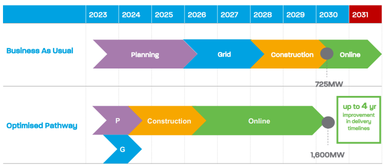
They explain the method somewhat. It is a linear regression across a number of economic forecasts, temperature and new anticipated technological load e.g. data centres.

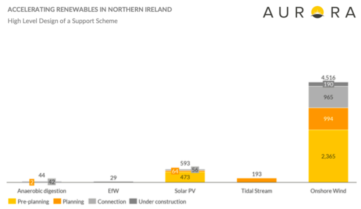
Requirement

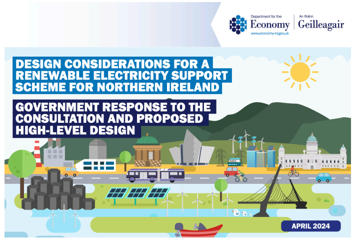
	Renewable	Estimate	TER	Estimate	Percentage	Notes
2025	3.1	Reported from NISRA	8	median scenario	41%	Current
2030	5.5	Our adjusted generation from SONI	10.2	median scenario	53%	Target Miss

Aurora Borrows heavily from a RenewableNI KPMG report

Lineage







RenewableNI

Aurora

DfE

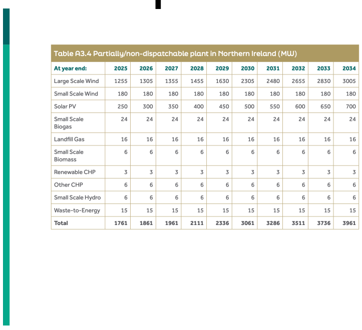


Table A3.4 Partially/non-dispatchable plant in Northern Ireland (MW)

At year end:	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
Large Scale Wind	1255	1305	1355	1455	1630	2305	2480	2655	2830	3005
Small Scale Wind	180	180	180	180	180	180	180	180	180	180
Solar PV	250	300	350	400	450	500	550	600	650	700
Small Scale Biogas	24	24	24	24	24	24	24	24	24	24
Landfill Gas	16	16	16	16	16	16	16	16	16	16
Small Scale Biomass	6	6	6	6	6	6	6	6	6	6
Renewable CHP	3	3	3	3	3	3	3	3	3	3
Other CHP	6	6	6	6	6	6	6	6	6	6
Small Scale Hydro	6	6	6	6	6	6	6	6	6	6
Waste-to-Energy	15	15	15	15	15	15	15	15	15	15
Total	**1761**	**1861**	**1961**	**2111**	**2336**	**3061**	**3286**	**3511**	**3736**	**3961**

All Island Resource Adequacy Assessment

Forecasts of Total Electricity Requirement

RenewableNI Report

This report lays the groundwork for the Aurora Report, which borrows heavily from it.

The renewableNI - KPMG report sets out a **primary route to market,** mainly the Contract for difference (CfD) scheme and notes that it has 7 years since a renewable support scheme has been available in Northern Ireland, and lack of newly commissioned renewables sources reflects the lack of support.

Only ~30 MW of new capacity was added in the past year, compared to 400 MW/year at the 2016 peak.

- Without intervention, NI will fall far short of 2030 targets — best-case under "business as usual" is 725 MW new capacity by 2030, less than 40% of what's needed.
- Grid, planning, and route-to-market issues are "single points of failure."

They describe baseline project development status

It outlines the stages of development projects are in and their aggregate generation and capacity, by stage and technology. They also note projects in the *pre-planning* stage, not in the REPD database.

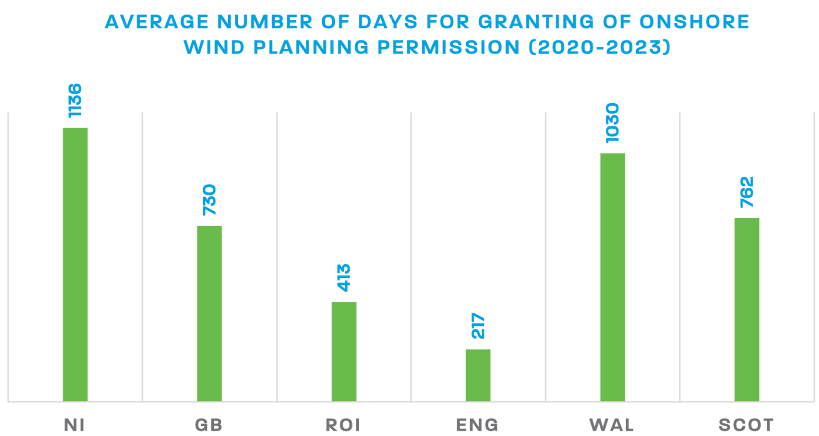
" What is apparent from this data is that it is highly unlikely the additional capacity required to achieve 2030 targets will be delivered from the existing pipeline"

They risk weight the success probability of projects in each stage reaching completion.

They describe planning, connection and construction lags

They benchmark project development timelines, giving estimates of the planning, connection and construction phases

Figure 3



BAU Development Pathway
Planning - 3 years
Grid application (6 months) and connection (18 months) - 2 years
Construction (18 - 24 months) - 2 years

Generation would only begin in 2030 (report is dated from 2023)
Capacity would only be **725 MW**

Optimised

Planning - 1 years
Grid application (6 months) and connection (18 months) - 2 years
OR Grid application (6 months) and expedited connection (6 months) - 1 year
OR **Concurrent** Planning permission with Grid application (6 months) and expedited connection (6 months) - 0 years
Construction (18 - 24 months) - 2 years - no change from BAU
Credible path to 2030 target
New capacity could be **1,600 MW**

See detailed breakdown below

Onshore Wind	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Additional capacity required by 2030 (MW)
Capacity	1577.25	43.8	149.49	1770.54	1000

Status	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Probability Weighting (business as usual)	Total Capacity delivered (MW)	Probability Weighting (optimised)	Total Capacity delivered (MW)
Under Construction	56.75	0	1.225	57.975	100%	57.98	100%	57.975
Consented with grid	186.3	0	0	186.3	80%	149.04	90%	167.67
Consented (no grid offer/unknown)	204.5	32.8	118.69	355.99	70%	249.19	80%	284.792
In planning	262.4	11	29.025	302.425	60%	181.46	70%	211.6975
Pre Planning	867.3	0	0.55	867.95	20%	173.59	40%	347.18
Total	1577.25	43.8	149.49	1770.54		811.25		1069.31

Solar	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Additional capacity required by 2030 (MW)
Capacity	673.6	16.85	0.2	690.7	400

Status	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Probability Weighting (business as usual)	Total Capacity delivered (MW)	Probability Weighting (optimised)	Total Capacity delivered (MW)
Under Construction	0	0	0	0	100%	0	100%	0
Consented	64.5	7.1	0	71.6	75%	53.7	90%	64.44
In planning	50	4.75	0.2	54.95	60%	32.97	70%	38.465
Pre Planning	559.1	5	0	564.1	20%	112.82	40%	225.64
Total	673.6	16.85	0.2	690.65		199.49		328.55

Offshore Wind	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Additional capacity required by 2030 (MW)
Capacity	1000	0	0	1000	500

	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Probability Weighting (business as usual)	Total Capacity delivered (MW)	Probability Weighting (optimised)	Total Capacity delivered (MW)
Pre Planning	1000	0	0	1000	10%	100	20%	200

In undertaking our delivery timeline assessment, we have used the following headline assumptions.

Onshore Wind	Probability Weighting (business as usual)	Assumptions	Probability Weighting (optimised)	Assumptions
Under Construction	100%	All projects currently under construction will be delivered and operational by 2030.	100%	All projects currently under construction will be delivered and operational by 2030.
Consented with grid	80%	20% of consented projects with grid will not be delivered on time due to route to market or grid delays.	90%	Reduced route to market risk with introduction of government backed subsidy scheme.
Consented (no grid offer/unknown)	70%	We have assumed a further 10% reduction in success probability to account for grid offer uncertainty.	80%	10% reduction in probability to account for grid offer uncertainty.
In planning	60%	DfI Renewable energy planning applications list, April 2017 (post NIRO) to March 2023, 63% of all renewable applications have been approved (246 / 391). The remainder have been rejected, undecided or withdrawn.	70%	10% increase in probability weighting due to reduced decision timelines, increased planning officer education and strict consultation parameters.
Pre Planning	20%	Fairly low success probability assumed due to early stage nature of the assets, with planning, grid and route to market risk all remaining high.	40%	Higher success rate assumed due to improved planning resources, grid investment and route to market options.

Solar	Probability Weighting (business as usual)	Assumptions	Probability Weighting (optimised)	Assumptions
Under Construction	100%	All projects currently under construction will be delivered and operational by 2030.	100%	All projects currently under construction will be delivered and operational by 2030.
Consented	75%	Same methodology as Onshore Wind but a blended rate has been applied due to lack of information on grid connection status.	90%	Reduced route to market risk with introduction of government backed subsidy scheme.
In planning	60%	DfI Renewable energy planning applications list, April 2017 (post NIRO) to March 2023, 63% of all renewable applications have been approved (246 / 391). The remainder have been rejected, undecided or withdrawn.	70%	10% increase in probability weighting due to reduced decision timelines, increased planning officer education and strict consultation parameters.
Pre Planning	20%	Fairly low success probability assumed due to early stage nature of the assets, with planning, grid and route to market risk all remaining high.	40%	Higher success rate assumed due to improved planning resources, grid investment and route to market options.

Offshore Wind	Probability Weighting (business as usual)	Assumptions	Probability Weighting (optimised)	Assumptions
Total	10%	Fairly low success probability assumed due to early stage nature of the assets, with planning, grid and route to market risk all remaining high.	20%	Higher success rate assumed due to improved planning resources, grid investment and route to market options

RenewableNI - Scenario Modelling

Business as Usual: 725 MW new capacity by 2030 → <40% to 80% target.

Optimised Pathway (if recommendations adopted): ~1,600 MW from existing pipeline + scope for additional projects → credible path to 80% target.

According to the 'Shaping Our Electricity Future Roadmap Version 1.1' published by EirGrid and SONI in July 2023, there is a need for Northern Ireland to have built a renewable generation capacity of **3,550 MW by 2030 in order to deliver the target** of 80% of electricity consumption from renewables sources.

Analysis suggests that to achieve this, **1,900MW** of new generation from onshore wind, solar and offshore wind will need to be added by 2030.

The NI market has a known pipeline of **3,461MW** of incremental renewables, from pre-planning to consented assets.

Assuming current timelines, and applying probability weightings, a best-case outcome of 725MW could commence generation by 2030. This is less than 40% of the volume required.

Furthermore, given the current timelines to progress a project from ideation to operation in NI, unless a project is already within the pipeline, it is highly unlikely to be able to contribute to the 2030 target.

This is outlined in the "Business as Usual" scenario.

The same pipeline of 3,461MW, but adopting and delivering upon the recommendations contained within this report, we estimate that the same pipeline could deliver 1,600 MW of generation capacity by 2030.

Furthermore, the enhanced timelines would provide an opportunity for further projects to be introduced into the system and still connect by 2030,

RenewableNI figures reported

3,550 MW	3,461MW	1,900 MW
Total RES to meet 80% by 2030	Known Pipeline	New RES needed added for 80% by 2030

Implies current RES baseline capacity is 1,650 MW
Assumes this baseline RES capacity stays until 2030

Table A3.5 All renewable energy sources in Northern Ireland (MW)										
At year end:	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**
All Wind	1435	1485	1535	1635	1810	2485	2660	2835	3010	3185
Solar PV	250	300	350	400	450	500	550	600	650	700
All Biomass/ Biogas/LFGas/ WTE	250	300	350	400	450	500	550	600	650	700
	79	79	79	79	79	79	79	79	79	79
Renewable CHP	3	3	3	3	3	3	3	3	3	3
Hydro	**6**	**6**	**6**	**6**	**6**	**6**	**6**	**6**	**6**	**6**

Table A1.1 Demand forecast for the median scenario in calendar year format for Ireland and Northern Ireland												
Median	**Calendar Year TER (TWh)**						**TER peak (GW)**			**Transmission peak (GW)**		
Year	**Ireland**		**Northern Ireland**		**All-Island**		**Ireland**	**Northern Ireland**	**All-Island**	**Ireland**	**Northern Ireland**	**All-Island**
	TER	**% growth**	**TER**	**% growth**	**TER**	**% growth**						
2023	34.0		8.09		42.1		5.76	1.6	7.36	5.64	1.57	7.21
2024	35.8	5.3%	8.21	1.6%	44.1	4.6%	5.95	1.59	7.54	5.83	1.56	7.39
2025	37.7	5.2%	8.33	1.4%	46.1	4.5%	6.13	1.61	7.72	6.02	1.57	7.57
2026	39.5	4.7%	8.57	2.8%	48.1	4.4%	6.32	1.67	7.97	6.21	1.63	7.82
2027	41.1	4.1%	9.05	5.7%	50.2	4.4%	6.46	1.75	8.19	6.35	1.71	8.04
2028	42.4	3.1%	9.31	2.9%	51.7	3.0%	6.57	1.79	8.33	6.46	1.76	8.18
2029	43.5	2.8%	9.58	2.9%	53.1	2.8%	6.64	1.84	8.45	6.53	1.80	8.30
2030	44.7	2.6%	9.83	2.5%	54.5	2.6%	6.72	1.87	8.53	6.61	1.84	8.39
2031	46.0	2.9%	10.11	2.9%	56.1	2.9%	6.83	1.91	8.66	6.71	1.88	8.52
2032	47.2	2.7%	10.47	3.5%	57.7	2.8%	6.92	1.95	8.79	6.81	1.91	8.63
2033	48.4	2.5%	10.71	2.3%	59.1	2.5%	7.03	1.99	8.90	6.91	1.96	8.76
2034	49.5	2.4%	10.94	2.2%	60.5	2.3%	7.12	2.04	9.03	7.01	2.00	8.88

Shaping Our Electricity Future Roadmap Version 1.1' published by EirGrid and SONI in July 2023

Table 12 provides a summary of the major renewable generation capacities assumed to be in place in Ireland and Northern Ireland by 2030.

Table 12: Summary of assumed renewable generation capacities in 2030

Source	Ireland (MW)	Northern Ireland (MW)
Onshore wind	9,000	2,450
Offshore wind	5,000 (+ 2,000 MW assigned to hydrogen production)	500
Solar[9]	8,000	600
Total	**22,000**	**3,550**

"The subsequent revised legal target of 80% RES-E in Northern Ireland by 2030 led to a recognition that offshore generation capacity would need to be delivered by 2030 to help achieve the target."

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Aurora Report

Commentary
Section 1.5 on page 17

1.5 Renewables pipeline in Northern Ireland
Since the closure of the Northern Ireland Renewable Obligation (NIRO) scheme for new entrants in 2017, no significant renewables capacity has become operational in Northern Ireland.
The renewables pipeline in Northern Ireland is dominated by onshore wind. There is good visibility on the capacity of plants having submitted planning applications, as they are listed in the UK government's Renewable Energy Planning Database (REPD). Onshore wind accounts for around 900 MW (76%) of capacity currently in the planning stage or beyond (grid connection and construction). Solar PV accounts for 13% with around 150 MW.

Offshore - targets from DfE of 1GW by 2030 are ambitious and not generally included.

The drivers of growth are **onshore wind and photovoltaics**

Construction Pipeline Progression

Success probability
100% progression from stage to stage

Time to Event of Pipeline Progression

> **Planning** 3 years
> **Grid connection** 2 years
> **Construction** 1 year
> Cumulative 6 years from planning app to Completion

Necessary Cadence of Progression to meet 80% by 2030

> **Planning** 1 year
> **Connection** 2 years
> **Construction** 1 year

Pipeline distribution (end-2023 snapshot)

By stage share of annual GWh in pipeline:

> Pre-planning 53%
> Planning 23%
> Connection 20%
> Under construction 4%

Aurora reference the AIRAA median demand forecast to 2030 and take 80% of this 10.2 TWh as the target.
Of this 8.2 TWh, 3.4TWh generation of renewables exist and are not going anywhere.
Of the remaining they estimate 1.5TWh (30%) are able to proceed with support from any scheme
This leaves 5TWh. Needed to be delivered by 2030 to be available.

Auction Design

Auctions are explained in the glossary in the appendix of the document. The economic and capacity constraints are not considered although the win rate, is.
Pot sizing and technology is not considered.

Aurora anticipate two auctions that impact the availability of new generation by 2030. the total sum of which is the 3.5GWh generation needed.

Auction 1: 2025 (delivery 2027) – 1,000 GWh/y (≈ 500 MW).

Auction 2: 2027 (delivery 2029) – 2,500 GWh/y (≈ 1,250 MW).

These are the **procurement ratios** Aurora says are needed
The "necessary conditions" case

Planning takes 1 year:

Auction 1 (2025): 48% of eligible volume must win.
Auction 2 (2027): 74% of eligible volume must win.

Planning takes 2 years

Auction 1: 85% must win (vs. 48%).
Auction 2: 92% must win (vs. 74%).

Both auctions could still meet volumes, but competition is much tighter and prices could rise.

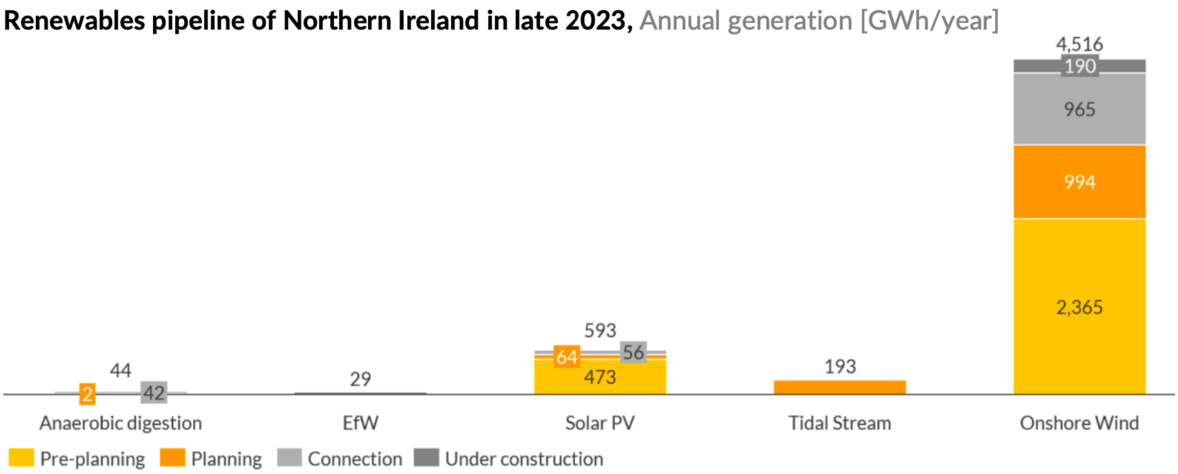
Other Assumptions

All capacity procured in Auctions 1 and 2 becomes operational on schedule (max 2 years after auction)

Projects delay construction to remain auction-eligible (e.g., 2024-eligible wait for 2025 auction, unsuccessful 2025 + 2026-eligible wait for 2027).

Pre-planning inflow assumed: extra 1,000 MW onshore + 600 MW solar; 50% get planning in 2026, 50% in 2027. This is in addition as to what is register in REPD and all of which will receive planning permission by the 2027 auction.

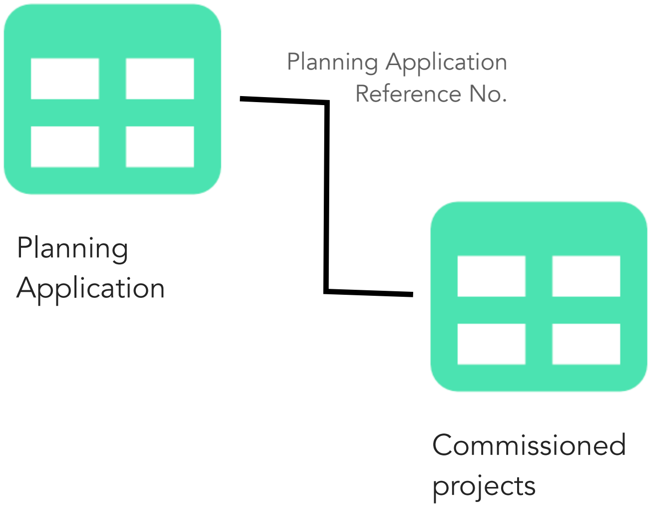
Current Pipeline



Sources: DESNZ REPD (planning stage and beyond); RenewableNI (pre-planning). See below for mapping reported project status to one of planning, connection, construction.

Our Modelling

Tracking projects across datasets



The NI Renewable Energy Planning Database (REPD) gives details of project planning applications and their status, whether, pending, granted, denied.

UK REPD gives Northern Ireland projects as well, but provides the fuller timeline of project development with timestamped dates of project submission, permission granted, construction started and project operational.

Categorising and Filtering Projects

We analysed the UK REPD to derive estimates of project progression success rate, to development time of project progression from stage to stage and to determine the baseline project development pipeline at which we will start our simulation.

Pipeline Status

We categorised viable projects as in the pipeline stages below, mapped to their assigned 'status'

Planning:

"Planning application submitted"
"Appeal lodged"
"Revised"

Connection:

“Planning Permission granted”
“Appeal granted”
“Secretary of state granted”

Construction:

“**Construction”**

as per Aurora criteria below

Process step	Description	Typical duration (years)
Planning	▪ Planning spans the period between a planning application being submitted and planning permission being granted ▪ Includes all projects with a status of ‘planning application submitted’, ‘appeal lodged’ and ‘revised’ in the REPD	1 – 3
Connection	▪ Connection spans the period from obtaining a grid connection to the beginning of the construction process ▪ Includes all projects with a status of ‘planning permission granted’, ‘appeal granted’, and ‘secretary of state granted’ in the REPD	1 – 2
Construction	▪ Connection spans the period from the start of construction to the COD and includes the period of financing the project ▪ Includes all projects with a status of under ‘construction’	1

Commissioned projects

Further Completed Projects (and not considered in the pipeline) :

“**Operational”**

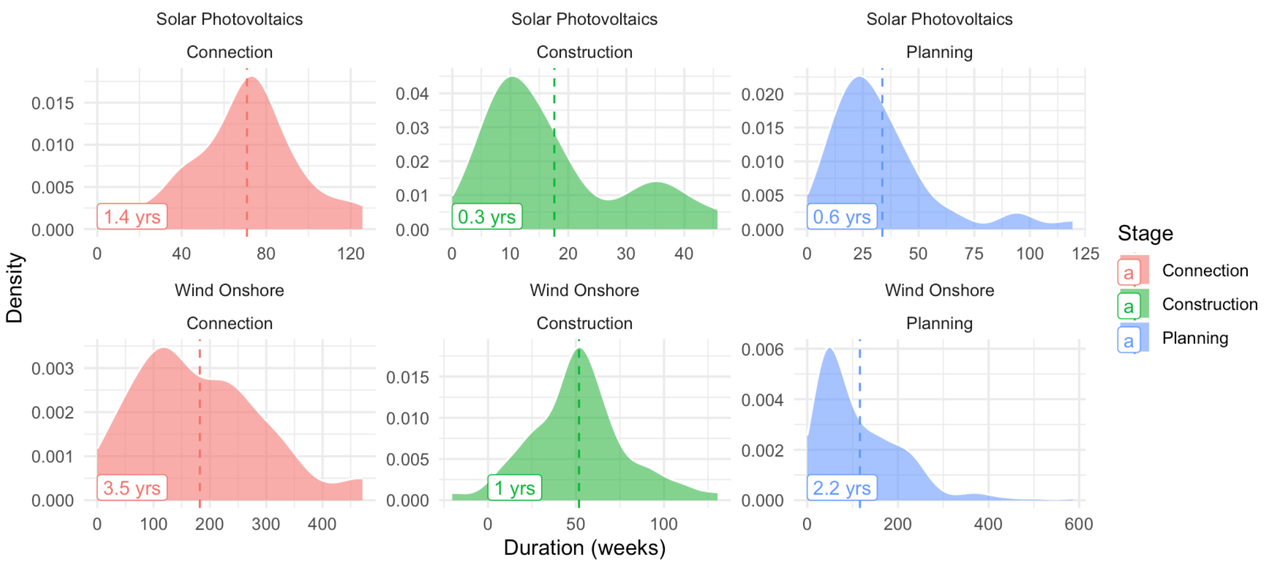
This gives our best guess at the pipeline of projects in development currently

Project development timelines

We look back at projects and take the end state of projects to determine how long that phase of development took projects using the below.

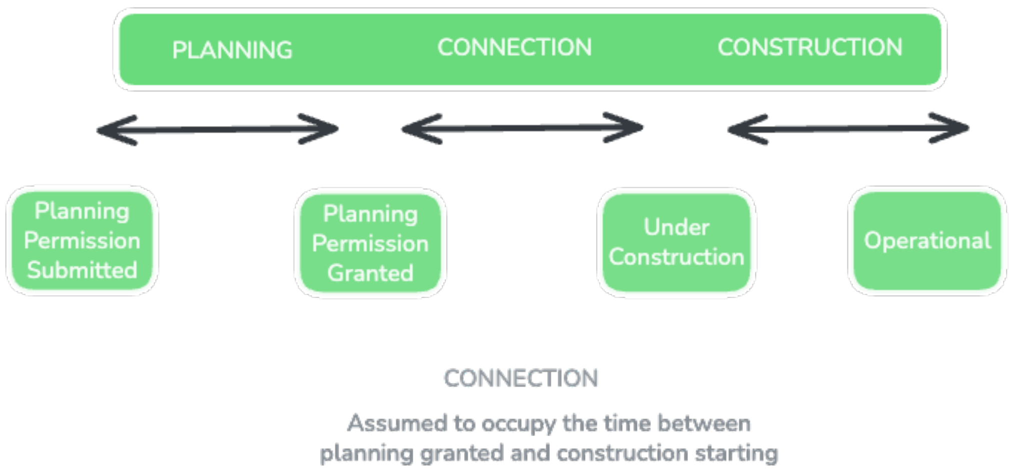
Time from Submission to Planning Granted, Connection finished and Operation (Construction Finished)

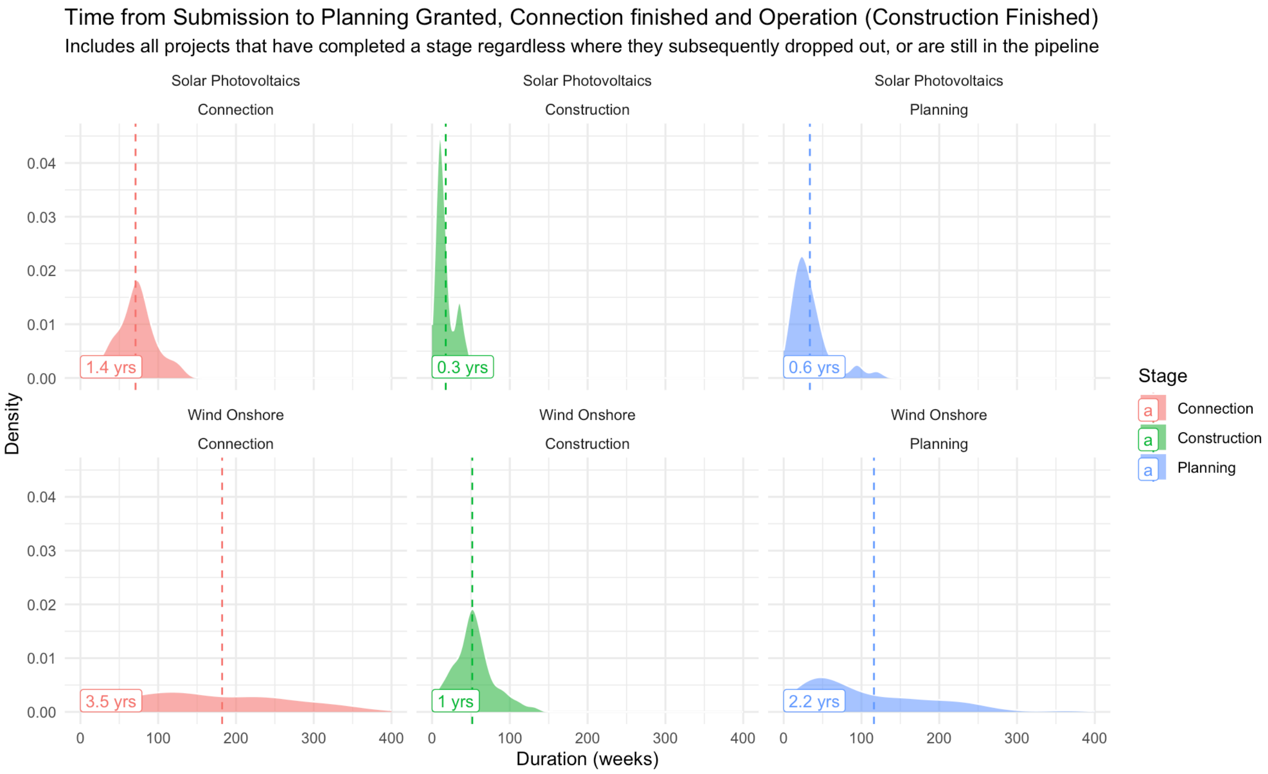
Includes all projects that have completed a stage regardless where they subsequently dropped out, or are still in the pipeline



Note: The scales on the x and y axis are free and not consistent

Planning = \`Planning Permission Granted\`- \`Planning Application Submitted\`
Connection = \`Under Construction\` - \`Planning Permission Granted\`
Construction = Operational - \`Under Construction\`





Note: The scales on the x and y axis are fixed for comparison

The same graph but with consistent x and y axis for comparison

Project development stage drop out rates

We look back at projects and note when a recorded `status` has indicated a project has not proceeded to the next stage.

#	Currrent Status	No Planning Granted Date (Planning phase)	No Construction Started Date (Connection Phase end)	No Operational Date (Construction Phase end)	N	Capacity (MW)
1	Completed	F	F	F	78	1083
2	Completed	F	F	T	1	23.5
3	Completed	F	T	F	22	279
4	Completed	F	T	T	1	0.2
5	Completed	T	F	F	1	18
6	Construction	F	F	T	1	27
7	Connection	F	T	T	119	483
8	Planning	F	T	T	6	115
9	Planning	T	T	T	96	439
10	Failed	F	T	T	23	152
11	Failed	T	T	T	73	988

Projects without Capacity estimates were assumed zero. These tended to be improvement projects rather than fresh new capacity. See below for imputation estimates using mean values.

We take projects that have 'revised' status and have been delivered a planning decision, as dropouts (**#9**). They will have started a new application. We track this instead. This prevents the planning duration estimate being unduly influenced by decision timelines that actually reflect multiple revisions to an application.

The number of projects that failed to proceed from the planning stage is 29, worth 1,103 MW. This is inclusive of revised projects.
The number of projects that failed to proceed from the Connection stage is 23, worth 152 MW.

Note some data quality issues are captured in the table.
#2 1 project was marked as operational despite having no operational timestamp
#4 1 project was marked as operational despite having no construction or operational timestamp.
#5 1 project was marked as operational despite having no planning granted date

Out of those 421 projects that are in the database, and minus 96 planning pending, there are 325. Of those 73 didn't leave planning/ progress from planning *for any reason.*

This gives a crude 73 / 325 ≈ 22.46% dropout or non progression rate for the planning stage

Out of those 252 projects that did proceed from planning, minus the 119 currently in the connection phase, leaves 132 projects. Including those 6 that subsequently revised their application and therefore didn't complete the connection phase, there are 29 projects that didn't complete connection *for any reason.*

This gives a crude 29 / 132 ≈ 21.97% dropout or non progression rate for the Connection stage

Of those 103 that completed the connection phase, 103 became operational. This implies the construction stage has a **0% dropout rate** or a **100% conversion rate**.

Note the milestone timestamp for moving between the connection phase and construction phase is the construction started timestamp. No connection finished or application success flag or timestamp is included in the REPD. A potential weakness of our analysis.

77.54%
Planning Success Rate

78.02%
Connection Success Rate

100%
Construction Success Rate

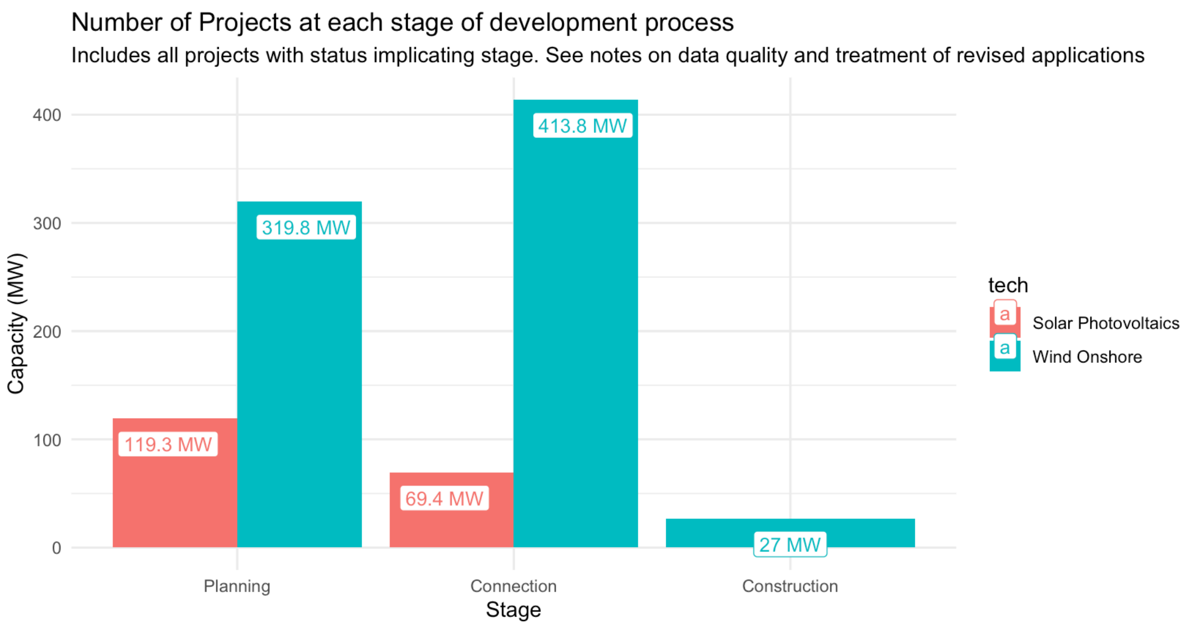
58.31%
Connection Success Rate
from Planning Stage

58.31%
Construction Success Rate
from Planning Stage

Current Project Development Pipeline

Tech	Currrent Status	N #	Capacity (MW)	Missing Capacity #	Average Capacity MW	Padded Capacity MW
Solar Photovoltaic	Completed	21	209	0	9.97	209
Solar Photovoltaic	Connection	11	69.4	5	11.6	127
Solar Photovoltaic	Planning	14	108	0	7.69	108
Solar Photovoltaic	Failed	18	119	4	9.18	156
Wind Onshore	Completed	82	1195	0	14.6	1195
Wind Onshore	Construction	1	27	0	27	27
Wind Onshore	Connection	108	414	9	4.18	451
Wind Onshore	Planning	79	320	30	6.53	516
Wind Onshore	Failed	88	1147	0	13	1147

Completed projects were discarded, as estimates of current renewable generation are given by NISRA.
The remaining pipeline projects are given by stage of planning and technology.

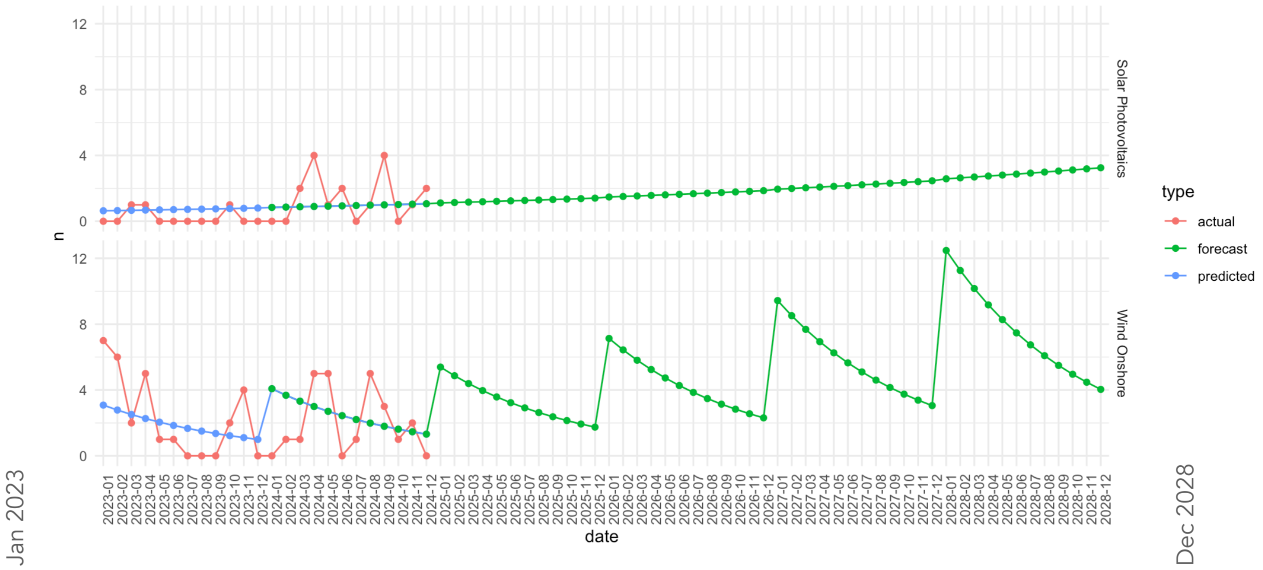


Pre - Planning

There is an acknowledgement that the current pipeline will be insufficient to meet the 80% by 2030 target

An estimate of those projects in the *pre-planning* phase, and not available in the planning database is necessary.

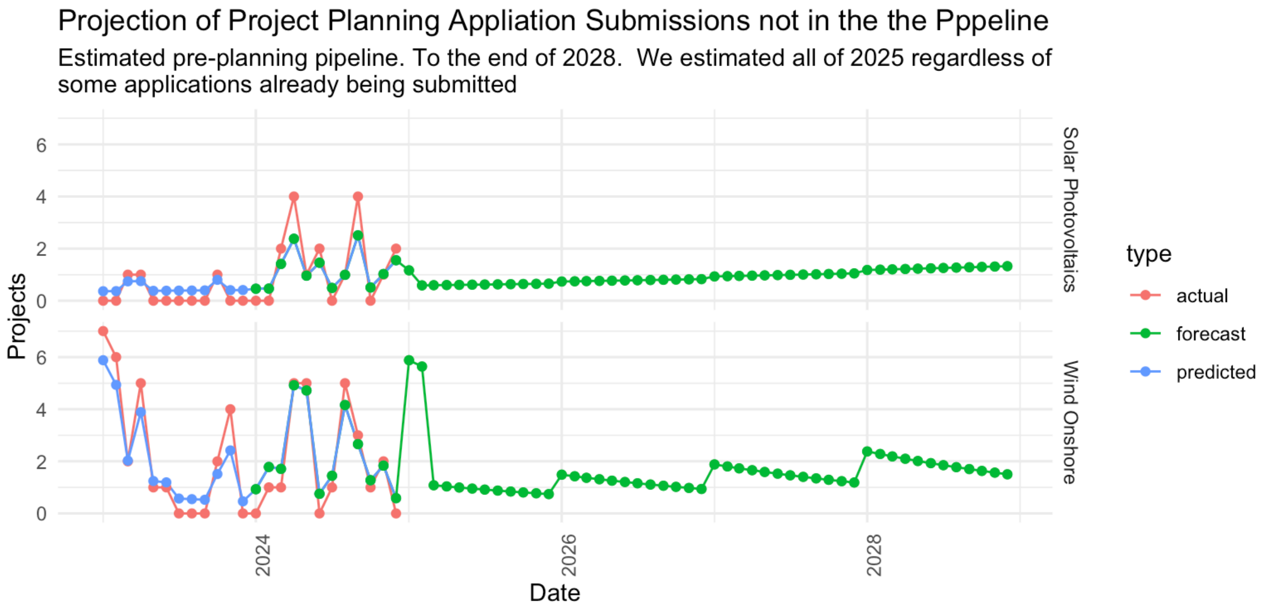
Very Optimistic Projections



*pd <- glm(data = data, formula = n ~ month*tech + year, family = 'quasipoisson')*

This shows a rapidly accelerating increase in project development rate (date is planning application submission) for both Solar Photovoltaic and Onshore Wind.
This is the most optimistic scenario for acceleration of renewables.

Reasonable Pre - planning Projections



*pd <- glm(data = data, formula = n ~ month*tech+year+offset(log(n+1)), family = 'quasipoisson')*

This shows a more conservative approach to estimating projects in pre-planning stage that will submit a planning application (plotted as Date in the x-axis). We treat this as business as usual. It shows a increasing trend year on year and is trained to anticipate the intra-year, monthly seasonality. It is also disaggregated by technology as above

This is the scenario we use below for 2030 estimation of renewable adoption.

Model

Results

These are the results for our base model. This took a conservative approach assuming business as usual. For tuning the input parameters see the next section.

We ran two models because of the slightly different treatment the development stages the existing pipeline was distribution across. We were able to assume that projected projects would all start from the planning stage.

So we projected projects in the pre-planning phase that we anticipate bringing forward a planning application at some time from start 2025 through to the end of 2028.

Statistical Notes

Skewed distribution of data required >> samples for bootstrapped confidence interval
Distributions of MW were right skewed
Distributions of MW were long tailed

Criticisms

If a project is in the pipeline already we take it to be at the start of the stage rather than at some point of progressing through the stage.

We may have double counted some early 2025 projects taking the current pipeline up to Q2 2025, and our projected projects from the start of 2025.

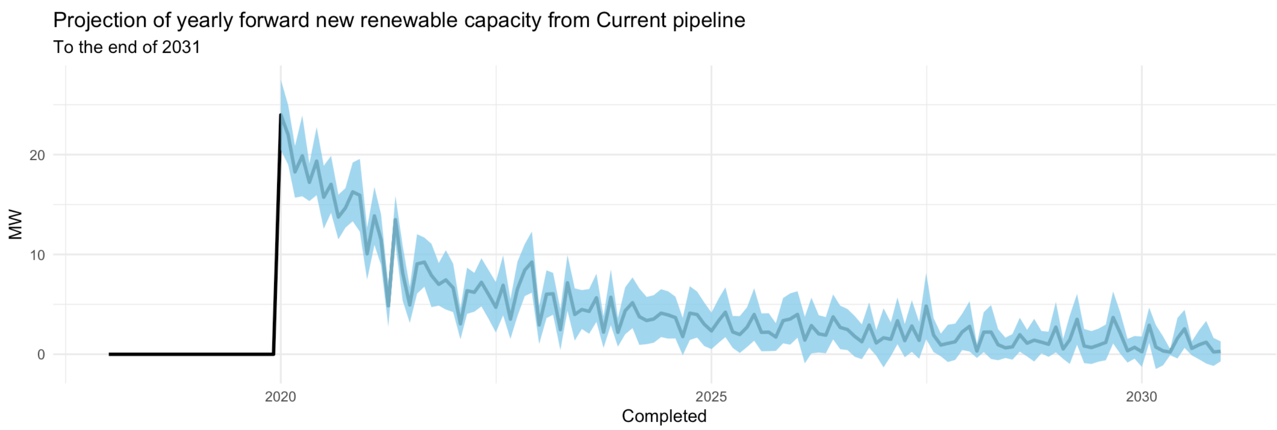
Data Quality

We rely on supplied data sets for data quality. Though we do note discrepancies.
Some projects are being reported as status: *awaiting construction*, despite their planning permission having expired and a other projects reporting a status as status: *planning expired* (e.g. *ref: 4251)*

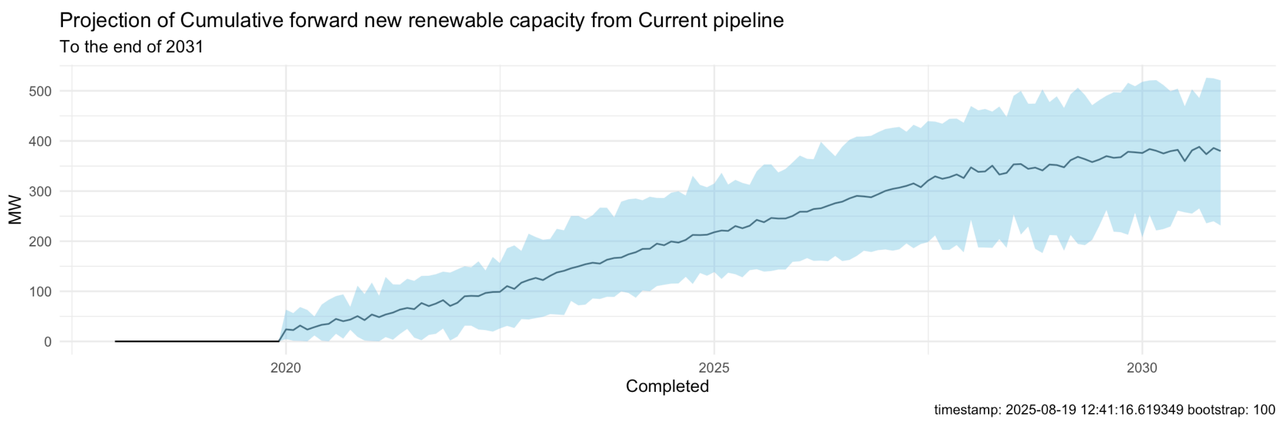
Some projects are missing Installed MW capacity figures. These were left blank as as manual inspection implied some of these were upgrade or refurbishment proposals with existing MW capacity and uncertain extra capacity.

Current Pipeline

Yearly



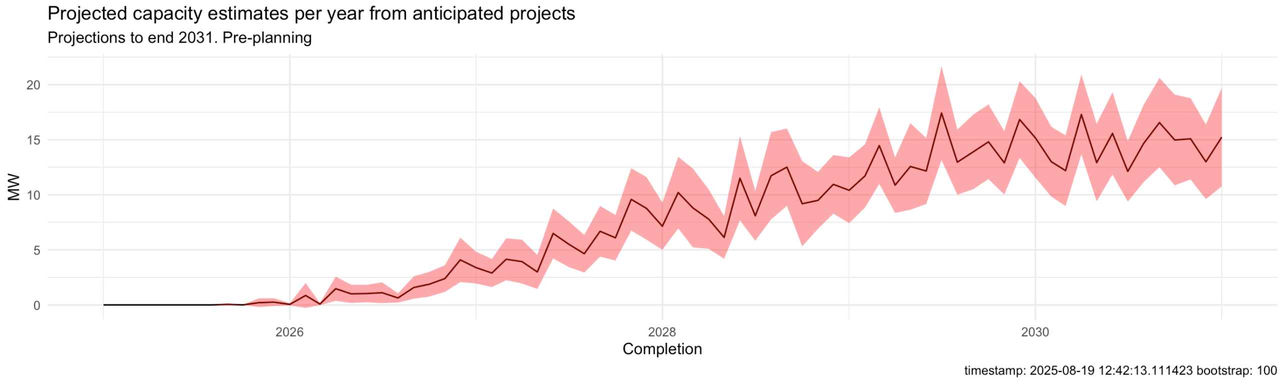
Cumulative



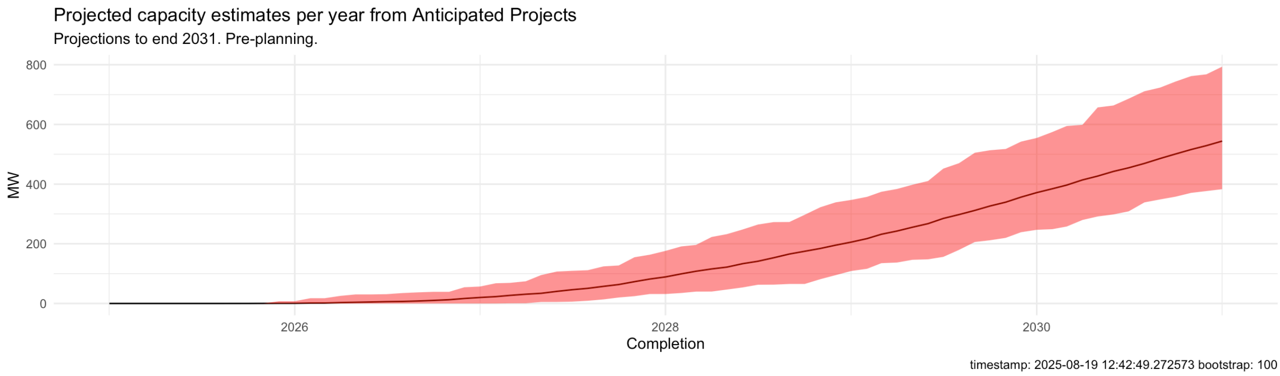
It is noted that projects in the current pipeline may have start dates before our current date due to the stochastic nature of our modelling. It is further noted that the these projects are substantial. Please consult the data quality notes.

Pre-planning Projects

Yearly



Cumulative



400 MW
New Capacity from Projected Pipeline
by 2030

400 MW
New Capacity from Current Pipeline
by 2030

800 MW

Total New Renewable Capacity by 2030

2460 MW

800 MW + 1640 MW

Total Renewable Capacity inclusive of current capacity by 2030

4151 MW

8 TWhr/yr

capacity factor of 22%

Total Capacity Requirement by 2030

60%

% Renewable by 2030

Target Miss

Interactive Application

Appendix

Data Sources

All Ireland Adequacy Assessment - SONI/ EirGrid

NISRA

Aurora Report

Solar photovoltaics deployment
Monthly deployment of all solar photovoltaic capacity in the United Kingdom.
https://www.gov.uk/government/statistics/solar-photovoltaics-deployment

Planning Portal
https://planningregister.planningsystemni.gov.uk/simple-search

Renewable Energy Planning Database - REPD - NI
List of planning applications for single wind turbines, wind farms and solar farms by status, April 2002 to March 2025.
https://www.infrastructure-ni.gov.uk/publications/renewable-energy-planning-applications-list-april-2002-march-2025

Renewable Energy Planning Database - REPD - UK
https://www.gov.uk/government/publications/renewable-energy-planning-database-monthly-extract

Commissioned Projects database

Resources

Climate Change Committee
https://www.theccc.org.uk/publication/adapting-to-climate-change-progress-in-northern-ireland/

Design considerations for a renewable electricity support scheme for Northern Ireland: response
https://www.economy-ni.gov.uk/publications/design-considerations-renewable-electricity-support-scheme-northern-ireland-response

Accelerating Renewables in Northern Ireland
KPMG - RenewableNI
https://renewableni.com/wp-content/uploads/2023/09/RNI-Report-Accelerating-renewables-in-Northern-Ireland-online-version.pdf

Aurora
Accelerating Renewables in Northern Ireland
A High level design of Renewable support scheme
https://www.economy-ni.gov.uk/sites/default/files/publications/economy/Aurora-accelerating-renewables-support-scheme-HLD.pdf

NIRO
https://www.economy-ni.gov.uk/sites/default/files/publications/economy/NIRO-how-it-works.pdf

DfE - road to net zero
https://www.economy-ni.gov.uk/publications/energy-strategy-path-net-zero-energy

DfE, NISRA Energy in Northern Ireland
https://www.economy-ni.gov.uk/sites/default/files/publications/economy/Energy-Northern-Ireland-2024-Report.pdf

References

NISRA
https://datavis.nisra.gov.uk/Economy/electricity-consumption-and-renewable-generation-report.html#2_Electricity_Consumption_from_Renewable_Sources_-_Monthly_data

Renewable Energy Support Scheme (RESS)
https://www.economy-ni.gov.uk/sites/default/files/publications/economy/Renewable-Electricity-Support-Scheme-response.pdf

KPMG RenewableNI - Economic Review of Small Scale Wind
Reports a capacity factor of ~22% for small scale wind
https://renewableni.com/wp-content/uploads/2021/06/An-Economic-Review-of-Small-Scale-Wind34.pdf
Undertaken for and on behalf of RenewableNI

Aurora
Page 80 reports a load factor of 23% on average for onshore renewable projects

Glossary

RES - Renewable Energy Source

RES-E stands for Renewable Energy Sources – Electricity. It is used to distinguish from other sources used for heat, or transport, RES-H and RES-T, for example.

Dispatchable Generation - Sources of electricity that can be used on demand and dispatched at the request of power grid operators according to needs
Does not include wind and solar generation which are non-dispatchable generation.

Capacity
GW is an energy per unit time measure. It is used to estimate ability to accommodate peaks in demand

Generation
Generation is referred to over time period. Typically a month or year and given in TWh.

Average GW from TWh/y

$$\textbf{GW (avg)} = \frac{\text{TWh per year}}{8.76}$$

(Leap year: divide by 8.784.)

Note the difference between nameplate and average measures of capacity and generation, below.

Capacity Factor

Generators will not produce at full power for the entirety of e.g. a year. To describe this a percentage metric between full and delivered capacity is calculated called a capacity factor.

For small scale wind installations this is estimates at 22%, meaning over some time period, such as a year the installation is only producing 22% of its possible (nameplate) output.

$$\text{Capacity Factor} = \frac{\text{Actual Energy Generated in period (MWh)}}{\text{Nameplate Capacity (MW)} \times \text{Hours in period}}$$

Load Factor is the same a capacity factor in the majority of cases

Nameplate capacity is the generation capacity under optimal conditions
If you want nameplate capacity at a capacity factor (CF):

$$\textbf{GW (nameplate)} = \frac{\text{TWh/yr}}{8.76 \times \text{CF}}$$

Example: 3.5 TWh/y @ 40% CF

$$\frac{3.5}{(8.76 \times 0.40)} \approx \mathbf{1.0\ GW}.$$

Contract for Difference
CfD - a government guaranteed contract providing a mechanism for a set price that electricity can be sold for.
The RESS is to be of this type.

Bid and Auctions
In the context for CfD bids are for a set strike price that sets the guaranteed electricity price.

Auctions are more typically referred to as **Allocation Rounds.**

Offers refer to the offered capacity e.g. 10GW.

Win Rate is the percentage of eligible energy volume that secures a contract

$$\textbf{Win Rate\%} = \frac{\text{Awarded GWh}}{\text{Eligible GWh}}$$

Procurement Ratio is the required win rate for some target, or other delivery set. Here generally referring to 80% by 2030.

$$\textbf{Procurement Ratio\%} = \frac{\text{Target GWh}}{\text{Eligible GWh}}$$

Competition Ratio or **Oversubscription ratio**

$$\textbf{Competition ratio} = \frac{\text{Eligible volume}}{\text{Awarded volume}} = \frac{1}{\text{Win rate}}$$

If you do not win a contract, you can resubmit given the project is still eligible. Otherwise the project is said to proceed **without support,** or is referred to as a merchant project.

Auctions are usually set a fixed capacity, or generation (MW or MWh) such as 1500 GWh/y over 10 years for example.

Alternatively they may by limited by a price or budget cap.

A smaller pot usually means more competition and higher prices

Auction Criteria for Project Eligibility
Capacity Threshold Typically > 5 MW
Planning Consent & Land Rights Required
Must have a valid Grid Connection Offer
Supply Chain Plan Required for ≥300 MW or floating offshore projects
Sealed bid, below admin strike price

Auction Pot
An auction may be divided into pots relevant to RES technologies or other.

Clearing Price
The highest winning bidding strike price that still fits within the pot's capacity.
Also said to be the price at which the auction *clears.*

It has units £/MWh.

Pay per clearing is a type of auction where all winning bidders get paid the auction's clearing price regardless of their individual bid.

An **Admin strike cap** is the ceiling placed on bids. Any projects bidding above this are disqualified.

NIRO - Northern Ireland Renewables Obligation
An initiative discontinued in 2017. Legislation for suppliers to present ROC (Renewable Obligation Certificates)

A **Corporate Power Purchase Agreement** (**CPPA**) is a long-term contract where a company buys electricity directly from a renewable energy generator, such as a wind or solar farm, rather than from a traditional electricity supplier. It is an example of an alternative route to market for renewable energy projects.

Figures

AIRAA

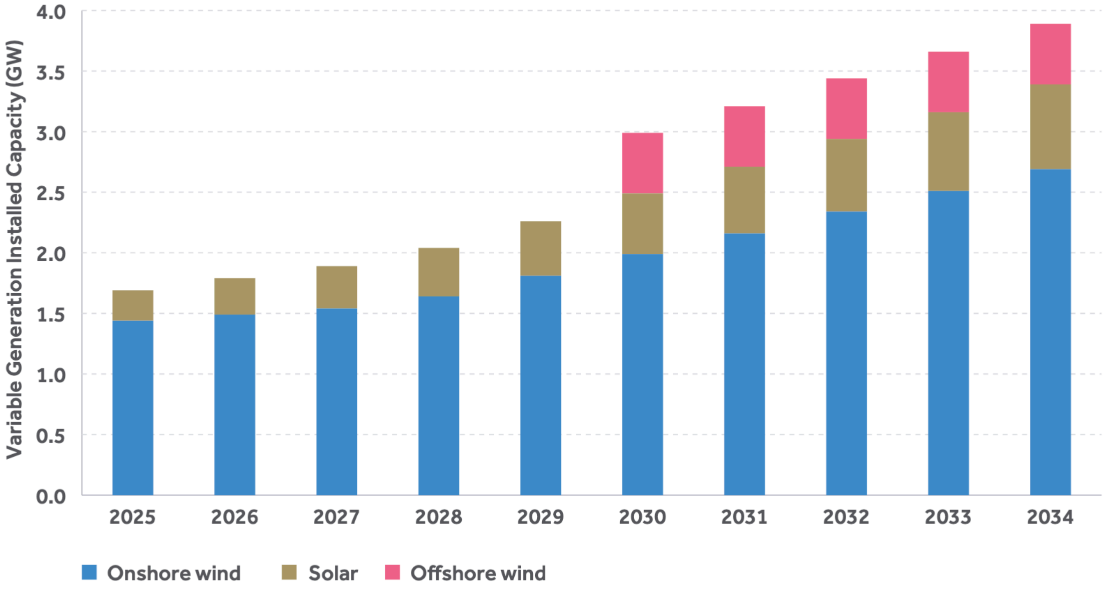


Figure 5.5 Assumed growth of renewable capacity in Northern Ireland

AURORA

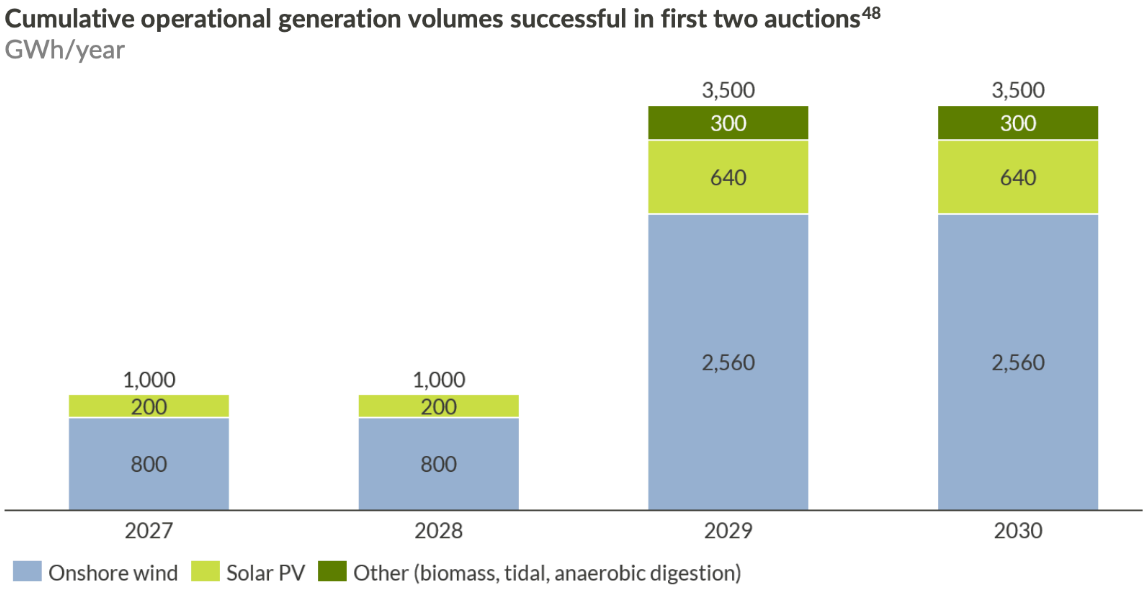
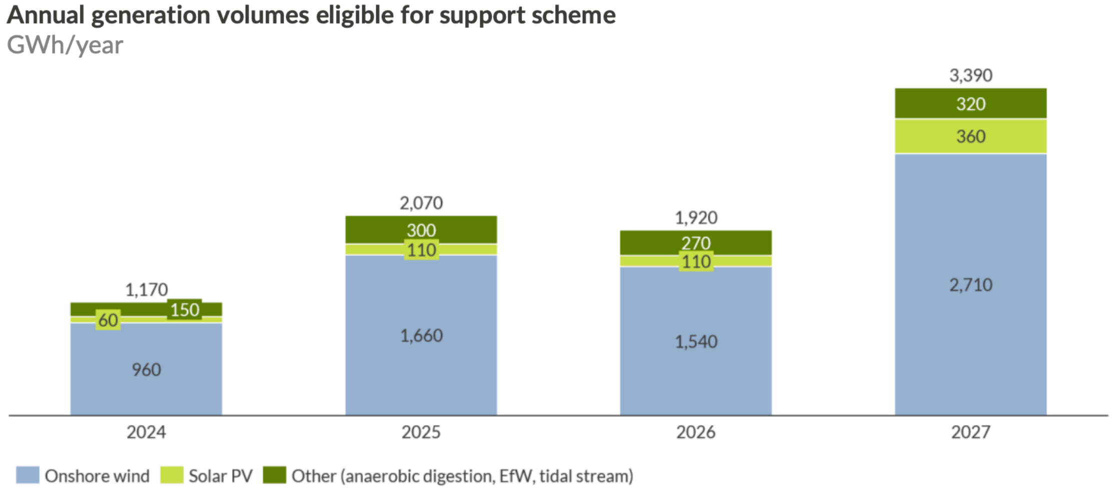


Figure 11: Cumulative operational generation volumes successful in first two auctions

Table 7 – Illustrative outcome of Auction 1

Pot	Technology	Generation (GWh/year)	Generation (% of total)	Capacity (MW)
1	Onshore wind	800	80%	340
1	Solar PV	200	20%	255



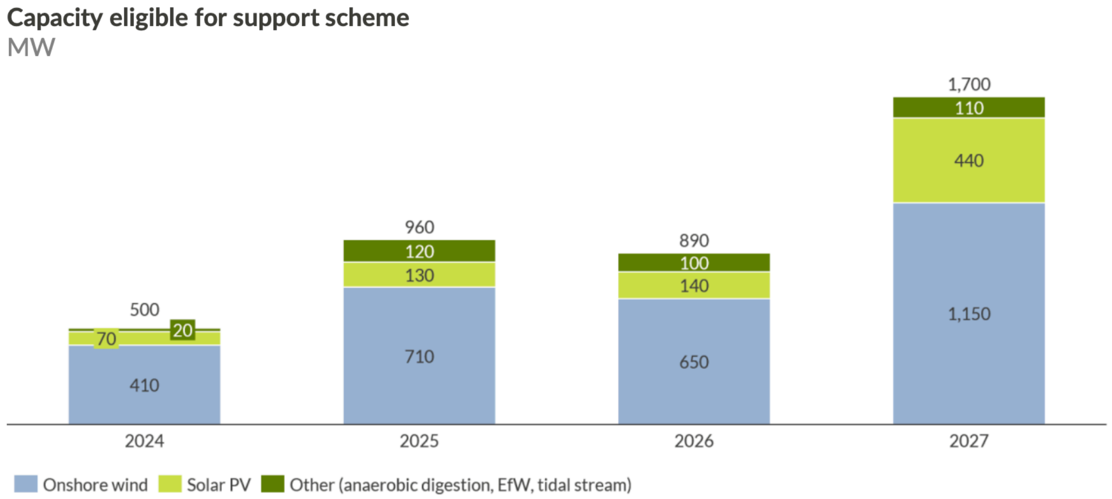


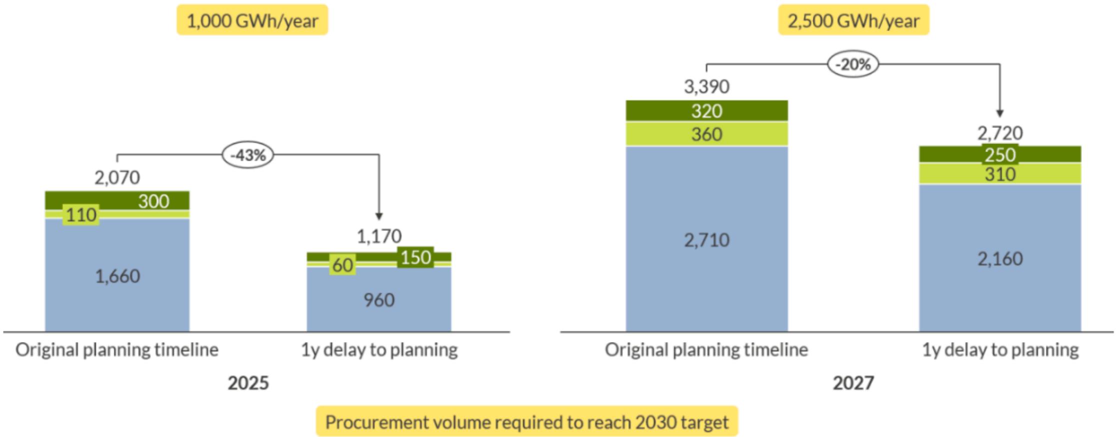
Figure 10: Onshore renewables capacity eligible for the Support Scheme (assuming 1 TWh is successful in 2025)

4.7 Sensitivity on planning timeline

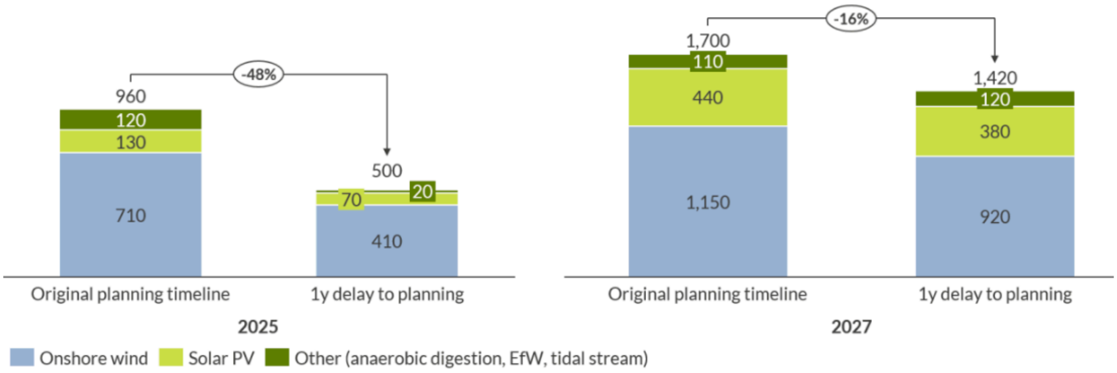
The below chart illustrates the effect of a one-year delay to planning timelines on the capacity and associated production volumes eligible for the Scheme[54] in the first auction (left) and second auction (right).

	Original planning timeline	1y delay to planning
Time for planning phase	1 year	2 years

Annual generation volumes eligible for support scheme, GWh/year



Capacity eligible for support scheme, MW



Under a one-year delay to the planning timeline, both auctions could still procure sufficient volumes, but with a much lower competition ratio. The first auction would need to procure 85% of eligible volumes (up from 48% in the one-year planning scenario); similarly, the second auction would need to procure 92% of eligible volumes (up from 74%). This could result in greater strategic bidding, and higher strike prices.